

Curriculum Vitae

Dr. Prem Singh Yadav
Principal Scientist

**Division of Animal Physiology & Reproduction,
ICAR-Central Institute for Research on Buffaloes
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Dr. Prem Singh Yadav

Father's Name: Sh. Sumer Singh

Date of birth & present age: 10.4.1963

Place of birth: Village Nimoth, Tehsil & District Rewari,
PIN -123411 (Haryana State) India.

Permanent home address: House Number 1425, Sector- 14 P
Hisar, 125001(Haryana State) India

Nationality: Indian

Correspondence Address: Principal Scientist
APR Division
ICAR-Central Institute for Research on
Buffaloes Hisar – 125001, Haryana (India).

Major achievements

- Dr PS Yadav (Team leader, ICAR Interdisciplinary Team Award), have been working on assisted reproductive technologies in buffalo including cloning, stem cells, IVF, OPU, ETT, cell banking, semen biology and transfer of technology
- In 2015, CIRB became India's second institute by producing a cloned buffalo bull, named Hisar-Gaurav (HG), In 2017, the team had cloned an Assamese buffalo bull in the field, away from the institute.
- Under my leadership ICAR-CIRB Hisar produced total 10 cloned buffalo calves during 2015-2020 and this achievement of producing maximum clones is recorded in Limca and India book of records. Produced 7 clones from a single superior bull M-29 are and re-cloned an earlier cloned bull Hisar-Gaurav.
- Produced >45000 semen doses from cloned bulls and > 5000 AIs from cloned bull semen, evaluated the performance of male and female progenies of cloned bulls resulting in increase in milk production and getting lakhs of doses of quality semen.
- Produced 1000 progenies from semen of cloned buffalo bulls.
- To initiate livestock stem cells in country including buffaloes and equines.
- Experience of recruitment of different categories of staff, purchase of animals, management of animals, management of calves, bulls management, training of bulls and semen production, creating housing facilities for bulls and heifers, instrument purchase, laboratory establishment, project formulation, project

- implementation, organizing workshops, training, brainstorming session, conferences and scientific meetings, etc
- Experience of institute building including agricultural farm management, farm machinery procurement and utilization, looking after works and waste management.
 - Five years' experience as Head of department during which different laboratories were developed and high numbers of national and international external funded project earned.
 - As team leader: NanaJi Deshmukh ICAR Team award for multidisciplinary research received
 - Completed 2 NASF- Mega projects of multi institutional, 4 DBT- Inter-institutional multidisciplinary, and 1 DAAD DST International project with Germany, 1- Network project having multi-institutional team and 17 Institute projects of total budgetary support of >20 crores.
 - Evaluated >100 research Projects for different funding agencies, awards and fellowships.
 - Completed 2 Post-Doc in Germany funded by DBT, India and DAAD, Germany
 - Invited as expert from China
 - Published 5 books, high impact factor peer reviewed research papers (>100), guided 10 students from 3 countries and five disciplines, published several compendiums, review articles, bulletins and delivered as expert in several conferences, workshops and trainings.
 - Experience of several committees as chairman, member secretary and members, such as IBSC, RAC, IMC, DBT task force for biotechnology and societal developmental.
 - Fellow of National Academy of Dairy Science, Fellow of Society of Animal Physiologist of India, Distinguish Scientist award of ISBD, Team Award of ISBD, Organizing secretary of SAPI, 2014

Academic record:

Degree/ Certificate	Year	Institution	Division	Subject
Bachelor's (BSc Hons. Agri.)	1985	HAU, Hisar Haryana	First	Agriculture, Anim. Sci.
Master's (MSc)	1987	HAU, Hisar Haryana	First	Animal Production

				Physiology
Doctorate (PhD)	1991	CCS HAU, Hisar Haryana	First	Animal Production Physiology
Post-Doc (DBT overseas Associateship)	Nov. 2003 to Nov. 2004	Institute of Animal Sciences Mariensee, 31535 Germany	-	Stem Cell in Bovines (Animal Biotechnology)
DAAD Research Stay	Dec.2010 -Feb 2011	Institute of Farm Animal Genetics, Mariensee, Holtstr 10, 31535 Germany	-	Induced Pluripotent Bovine Stem Cells

Post Doc Germany Project DAAD Title **Induced Pluripotent Bovine Stem Cells**
 Post Doc Germany Project DBT Overseas Title **“Study developmental potential of
 bovine embryonic stem Cells”**

Ph.D. Thesis title: **“In vitro maturation and fertilization of goat oocytes”.**

M.Sc. Thesis title: **“Age influenced physiological changes in male Japanese quails”.**

Professional Record:

<i>Position</i>	<i>Institute</i>	<i>Period</i>	
		<i>From</i>	<i>To</i>
Prof cum Principal Scientist	CIRB Hisar	2.2.2008	<i>Continue</i>
Senior Scientist	CIRB Hisar	2.2.2000	1.2.2008
Scientist (Sr. Scale)	CIRB Hisar	12.4.1998	2.2.2000
Scientist	NAARM Hyderabad, IVRI Izatnagar & CIRB Hisar,	12.4.1993	11.4.1998

Training Obtained:

1. Management Development Program on Leadership Development (Pre-RMP) 15-26 July 2014
2. Training on ISO 9001 May 2-5, 2013
3. Workshop on Stress management – Medical angle at NAARM February 12-14, 2013
4. Strengthening statistical computing for NARS- SAS training at CIRB Hisar
5. Management Development Program on Public Private Partnership, July 20-24, 2009” at IIM, Lucknow
6. Program on “Scientific project formulation implementation and evaluation” 17-28 March, 2008 Administrative Staff college of India, Hyderabad
7. Somatic cell banking and its application. From 28.4.2003-3.5.2003 at NBAGR Karnal.

8. Advances in Embryo Biotechnology” From December 22, 2001 to January 11,2002 at GBPUAT Pantnagar-263145 INDIA
9. Web Page Designing and Computer Applications” From February 6-27, 2000 at IASRI New Delhi.110012 INDIA
10. Training in Animal production and Management”. April 8-25, 1996 at CIRB Hisar-125001 INDIA
11. Foundation course on research management at NAARM Hyderabad 12 April to 16 September 1993

Membership of Scientific Societies:

1. Indian Science congress- **Life member**
2. Society of Animal Physiologists of India – **Life Member**
3. Indian Society for Study of Animal Reproduction- **Life member**
4. Indian Society for Buffalo Development- **Life member**
5. Indian Society for Study of Reproduction and Fertility- **Life Member**
6. Indian Society for Conservation of Animal Biodiversity- **Life member**

Reviewer of peer reviewed journal

1. Jury member, 12th Royan International Research Awards, Tehran – Iran 2011
2. Stem Cell and Development
3. Cytotechnology
4. Sustainable Agricultural Research , Canadian center of science and education
5. Nutrition Reviews
6. Indian Journal of Animal Science
7. Indian Journal of Agricultural Research
8. Indian Journal of Biotechnology
9. DBT Projects

Scholarships / Fellowships / Awards / Professional recognition:

1. Expert in DBT Brain Storming on Reproductive Biotechnologies 21-22 March 2016 at AAU Khannapara Guwhati
2. Special Invitee, Task force DBT 16-17 November, 2015
3. Expert in DBT Workshop on Reproductive Biotechnologies for enhancement of livestock productivity 20-21 January 2014 at NIAB, Hyderabad
4. Course Director DBT Sponsored short course 2014
5. DBT funded project on “Isolation and Characterization of Mesenchymal stem cells from amniotic fluid. Funding of Rs 22.85 Lakhs 2012
6. Jury member, 12th Royan International Research Awards, Tehran – Iran 2011
7. DAAD research stay and study visit 2010 to Germany
8. Best Poster Award in SOCDAB Conference 2010 at Anand Agricultural University , Anand, Gujrat Feb 10-11, 2010
9. 2nd Best Poster Award in ICB 2012 at CDLU, Sirsa Sept 18-20, 2012
10. DBT funded project on buffalo adult stem cells with a funding of Rs 67.13 Lakhs 2007-2011
11. Course Director- ICAR sponsored short course
12. Post Doc Fellowship- Long term DBT Overseas Associateship in Germany for the year 2002 -2003.

13. Outstanding Young Scientist award by the Indian Society for Study of Animal Reproduction in the year 1996-1997.
14. ICAR Sr. Fellowship for Ph. D.
15. HAU Merit Scholarship for MSC.

Students Guided

Sr. No	Student	Degree	University	Year	Thesis Title
1	Dr. Gulam Ali	PhD	A P P, CCSHAU Hisar	2000-2002	Micromanipulation of buffalo embryos for sexing using PCR
2	Dr. Yousef Denke	MVSc	Animal Biotech. CCSHAU Hisar	2006-2007	Vitrification of buffalo oocytes and their post-thaw potential for in vitro fertilization.
3	Jarnail Singh	PhD	Biotechnology CDLU Sirsa	2009-12	Isolation, culture and characterization of umbilical cord stem cells from buffaloes
4	Dr. Priti Singh	MVSc	Vety Physiology CCSHAU Hisar	2009-2011	Umbilical cord stem cell freezing and apoptosis
5	Dr. Manisha	MVSc	IVRI, Izatnagar	2012-2013	Identification of osteopontin and clusterin in buffalo semen and their relation to semen quality
6	Dr. Fozia Shah	PhD	Vety Physiology HAU Hisar	2010-14	Studies on donor cells for production of cloned embryos and their characterization in buffalo (<i>Bubalis bubalis</i>).
7	Dr. Sadeesh EM	PhD	Biochemistry IVRI, Izatnagar	2012-14	Production of cloned buffalo embryos and expression profile of developmentally important genes during embryogenesis
8	Kaushalya	PhD	Biotechnology CDLU Sirsa	2012-2015	Studies on stemness properties of cultured buffalo amniotic membrane cells
9	Dr. Papori Sharma	PhD	Animal Genetics LUVAS Hisar	2014-2016	Studies on epigenetic characteristics of somatic cells in water buffaloes
10	Dr MA Wakil	PhD	Animal Physiology)		

Conference/ Short Course/ Summer Schools organized

1. Annual Conference of Society of Animal Physiologists of India (SAPI 2014) and National Symposium on Physiological determinants of climate resilient and sustainable animal production as Organizing Secretary Nov 27-28 2014 at ICAR-CIRB Hisar.

2. DBT Sponsored short course on Reproductive biotechnology – an integrated approach to improve reproductive efficiency. 11-26 March 2014 (As short course Director)
3. Agri-education day 2012 , 400 school students from various schools 27.11.2012. . (Coordinator)
4. Application of embryonic stem cells in livestock research and production December 10-19, 2007 (As Director Short Course)
5. Artificial breeding and reproduction management in buffaloes, June 10-30, 2003. (Co-coordinator -Summer School)
6. Application of cryopreservation of buffalo germplasm for improving productivity of dairy buffaloes, June 1-21, 2001. (Co- coordinator- Summer School)

Book Authored / Edited

1. PS Yadav, Pradeep Kumar and Jerome A Eds (2014) Reproductive biotechnology – an integrated approach to improve reproductive efficiency. CIRB Publication ISBN 978-93-6156-440-9 pp 1-224
2. Pradeep Kumar, P S Yadav and RK Sethi Applied Veterinary Andrology and Artificial Insemination Kalyani publications 2013 ISBN 978-93-272-2982-0
3. RK Sharma, SK phulia , Inderjeet Singh and PS Yadav Kritrim Garbhadhan and Bhains Palan 2012. Hindi CIRB, Publication
4. P S Yadav B Singh, I. Singh and RK Sethi 2010 Reproductive Biotechnology in Buffalo. Satish Serial Publishers. Delhi India ISBN 81-89304-80-1 PP192
5. **Asian Buffalo Magazine, Vol 9 No2, July-Dec 2015.pp28**
6. **Training Compendium:** Ultrasonography in large animal reproduction for fertility augmentation. RK Sharma , SK Phulia, A. Jerome, S Khanna and PS Yadav pp165

Book Chapters

1. P S Yadav et al 2014 Reproductive biotechnology – an integrated approach to improve reproductive efficiency. Reproductive biotechnology – an integrated approach to improve reproductive efficiency. ISBN 978-93-6156-440-9 pp1-18.
2. B Singh PS Yadav and AK Rawat 2014 The state of the art for genomic reprogramming and induced pluripotency in livestock. Reproductive biotechnology – an integrated approach to improve reproductive efficiency. ISBN 978-93-6156-440-9 pp 32-45.
3. B Singh, G Mal, PS Yadav and AK Rawat 2014 Stem cell research to advance livestock sector. Reproductive biotechnology – an integrated approach to improve reproductive efficiency. ISBN 978-93-6156-440-9 pp 55-68.
4. Selokar NL, MS Chauhan, SK Singla, and PSYadav 2014 Genetic modified livestock animals through transgenic technology Reproductive biotechnology – an integrated approach to improve reproductive efficiency. ISBN 978-93-6156-440-9 pp82-87.
5. Varij Nayan, Anuradha Bhardwaj, and PS Yadav 2014 Indispensable culture considerations during in vitro embryo production for efficient reproductive

- biotechnology. Reproductive biotechnology – an integrated approach to improve reproductive efficiency. ISBN 978-93-6156-440-9 pp 88-100
6. BS Punia, PS Yadav and Birbal Singh 2014 Nutritional strategies for efficient reproduction Reproductive biotechnology – an integrated approach to improve reproductive efficiency. ISBN 978-93-6156-440-9 pp 144-148.
 7. PS Yadav 2014 Requirements, media preparation and cell culture procedures. Reproductive biotechnology – an integrated approach to improve reproductive efficiency. ISBN 978-93-6156-440-9 pp 175-189
 8. P S Yadav and BR Gulati 2014 Fetal mesenchymal stem cells in farm animals – Application in health and production. Biotechnology: Prospects and Applications ISBN 978-81-322-1682-7
 9. D Kumar and PS Yadav 2013 Future prospectus of stem cells in intervention in livestock reproduction. Physiology and Nutrigenomics underpinning animal production. Pp 59-65 ISBN -13:81-928693-1-5
 10. B. Singh, M Mukesh, SK Gautam M. V. Verma, SP Yadav and P S Yadav 2010 Post Genomic Era of Biology and Medicine: A Perspective on Livestock Production and management. Advances in Medicine and Biology Vol. 16 Biomedical Press, Nowa Science Publishers USA
 11. B. Singh, SK Gautam, MS Chauhan, SK Singla, S. Kumar, V Kumar and P S Yadav 2010 Cellular Reprogramming –Advances and Opportunities for Application in Veterinary and Animal Sciences, Advances in Medicine and Biology Vol. 16 Biomedical Press, Nowa Science Publishers USA
 12. B. Singh, M Mukesh, M. Sodhi, SK Gautam M. Kumar and P SYadav 2009 Nutrigenomics, metagenomics and metabolonomics: emerging faces of molecular genomics and nutrition. Nowa Science Publishers ISBN:978-1-61668-006-0 USA
 13. P S Yadav, R K Sharma and Pawan Singh 2006 Jav Prodyogiki me Bhains Ka yogdan and Sambhavanaya pp128-134 Bhains Utpadan and Parbahndhan CIRB publication.
 14. Pawan Singh , Anurag Bhardwaj and P S Yadav 2006 Sand Ka Chayan Parbhandhan and Upyog pp 86-91 Bhains Utpadan and Parbahndhan CIRB publication.

Compilations/ Editing's

1. RK Sharma PS Yadav Ultrasonography compendium 2016
2. PS Yadav, Jerome A, Kumar P , Naresh Seloakr , Phulia SK and Kumar D. 2014 Compendium abstracts and national symposium papers on Physiological determinants of climate resilient and sustainable animal production. IIIrd annual conference of SAPI, Nov 27-28 2014 CIRB, Hisar
3. SS Dahiya, PS Yadav and RK Sethi 2011. Vision 2050, Central Institute for Research on Buffaloes, Hisar- 125001
4. SS Dahiya, PS Yadav, N Rana, D Kumar and RK Sethi 2011. Vision 2030, Central Institute for Research on Buffaloes, Hisar- 125001
5. PS Yadav, Inderjeet Singh, Pawan Singh, RK Sharma and SK Phulia 2007 Application of embryonic stem cells in livestock research and production December 10-19 2007

6. Inderjeet Singh, P S Yadav and RK Sharma 2003 Artificial breeding and reproduction management in buffaloes, June 10-30, 2003.
7. Inderjeet Singh, PS Yadav Proceedings of Asian Buffalo Congress 2003
8. P S Yadav, Inderjeet Singh, OK Hooda and Sajjan Singh 2001 Application of cryopreservation of buffalo germplasm for improving productivity of dairy buffaloes, Compendium of Summer School, June 1-21, 2001.
9. International buffalo congress
10. Asian buffalo congress

Training Manuals

1. PS Yadav, RK Sharma, Pawan Singh S K Phulia and AK Bhalara 2007 Practical Manual "Application of embryonic stem cells in livestock research and production" December 10-19 2007 CIRB Hisar Publication
2. PS Yadav, RK Sharma, Pawan Singh, JK Singh and Inderjeet Singh 2008 A Manual on in Vitro Production of livestock embryos. CIRB Hisar Publication

Publications

1. Punetha M, Kumar D, Saini S, Chaudhary S, Bajwa KK, Sharma S, Mangal M, **Yadav PS**, Green JA, Whitworth K, et al. 2024. Optimising electroporation condition for CRISPR/Cas-mediated knockout in zona-intact buffalo zygotes. *Animals*. 14(1):134. <https://doi.org/10.3390/ani14010134>
2. Sharma M, Punetha M, Saini S, Chaudhary S, Jinagal S, Thakur S, Kumar P, Kumar R, Sharma RK, **Yadav PS**, Kumar D. 2024. Mito-Q supplementation of in vitro maturation or in vitro culture medium improves maturation of buffalo oocytes and developmental competence of cloned embryos by reducing ROS production. *Anim Reprod Sci.*; 260:107382. doi: 10.1016/j.anireprosci.2023.107382.
3. **Yadav PS**, Kumar D, Saini M, Sharma RK, Dua S, Selokar NL, Bansal S, Punetha M, Gupta A, Kumar R, Kumar P. 2024. Evaluation of postnatal growth, hematology, telomere length and semen attributes of multiple clones and re-clone of superior buffalo breeding bulls. *Theriogenology*; 213:24-33. doi: 10.1016/j.theriogenology.2023.09.024.
4. Bajwa KK, Punetha M, Kumar D, **Yadav PS**, Long CR, Selokar NL. 2023. Electroporation-based CRISPR gene editing in adult buffalo fibroblast cells. *Anim Biotechnol.*;1-12. doi: 10.1080/10495398.2023.2271030.
5. Dalal J, Kumar P, Chandolia RK, Pawaria S, Bala R, Kumar D, **Yadav PS**. 2023. A new role of H89: Reduces capacitation-like changes through inhibition of cholesterol efflux, calcium influx, and proteins tyrosine phosphorylation during sperm cryopreservation in buffalo. *Theriogenology*; 204:31-39. doi: 10.1016/j.theriogenology.2023.04.001.
6. Dua S, Bansal S, Gautam D, Jose B, Singh P, Singh MK, De S, Kumar D, **Yadav PS**, Kues W, Selokar NL. 2023. Production of MSTN gene-edited embryos of

- buffalo using the CRISPR/Cas9 system and SCNT. *Cell Reprogram*; 25(3):121-127. doi: 10.1089/cell.2023.0003.
7. Punetha M, Saini S, Chaudhary S, Bala R, Sharma M, Kumar P, **Kumar D, Yadav PS**. 2023. Mitochondria-targeted antioxidant MitoQ ameliorates ROS production and improves cell viability in cryopreserved buffalo fibroblasts. *Tissue Cell*; 82:102067. doi: 10.1016/j.tice.2023.102067.
 8. Arjun V, Kumar P, Dutt R, Kumar A, Bala R, Verma N, Jerome A, Virmani M, Patil CS, Bhardwaj S, Kumar D, **Yadav PS**. 2022. Effect of mitochondria-targeted antioxidant on the regulation of the mitochondrial function of sperm during cryopreservation. *Andrologia*; 54(7):e14431. doi: 10.1111/and.14431.
 9. Dua S, Sharma P, Saini M, Rawat N, Rajendran R, Bansal S, Wakil AM, Beniwal M, Parashar A, Bajwa KK, Selokar NL, Kumar R, Kumar D, **Yadav PS**. 2021. Cryobanking of primary somatic cells of elite farm animals - a pilot study in domesticated water buffalo (*Bubalus bubalis*). *Cryobiology*, 98:139-145. <https://doi.org/10.1016/j.cryobiol.2020.11.014>.
 10. Saini M, Sheoran S, Vijayalakshmy K, Rajendran R, Kumar D, Kumar P, Sharma RK, Kumar R, Mohnaty TK, Selokar NL, **Yadav PS**. 2020. Semen parameters and fertility potency of a cloned water buffalo (*Bubalus bubalis*) bull produced from a semen-derived epithelial cell. *PloS one*, 15(8), e0237766. <https://doi.org/10.1371/journal.pone.0237766>
 11. Dalal J, Chandolia RK, Pawaria S, Kumar A, Kumar D, Selokar NL, Andonissamy J, **Yadav PS**, Kumar P. 2020. Low-density lipoproteins protect sperm during cryopreservation in buffalo: Unravelling mechanism of action. *Molecular Reproduction and Development*, 87(12); 1231-1244. <https://doi.org/10.1002/mrd.23434>.
 12. Rajendran R, Saini M, Dua S, Saini D, Kumar D, **Yadav P S**, Selokar NL. 2019. Simple somatic cell assay to screen mammalian embryo toxicity caused by mineral oil. *Current Science*, 117 (8): 1270-1271.
 13. Kumar P, Pawaria S, Dalal J, Ravesh S, Bharadwaj S, Jerome A, Kumar D, Jan M H, **Yadav P S**. 2019. Sodium alginate potentiates antioxidants, cryoprotection and antibacterial activities of egg yolk extender during semen cryopreservation in buffalo. *Animal Reproduction Science*, 209: 106166. <https://doi.org/10.1016/j.anireprosci.2019.106166>
 14. Saini M, Selokar N, Rajendran R, Kumar D, Kumar P, **Yadav PS**. 2019. Isolation and culture of epithelial cells from stored buffalo semen and production of cloned embryos from semen-derived epithelial cells. *Reproduction, Fertility and Development*, 31(10): 1581-1588. <https://doi.org/10.1071/RD18356>.
 15. Elzein E, Osman I, Sharma RK, Badawi ME, Dahiya S, **Prem Yadav**, Phulia A, Kumar D, Selokar N, Jerome A, Kumar P, Saini M, Rawat Nidhi, Wakil A, Rasika R. 2019. Influence of different stages of corpus luteum on ovary size, oocytes grades

- and follicular population in Indian buffaloes. International Journal of Environment, Agriculture and Biotechnology (IJEAB), 4(5): 1356-1359.
16. Kumar D, Anand T, Vijayalakshmy K, Sharma P, Rajendran R, Selokar NL, **Yadav PS**, Kumar D, 2019. Transposon mediated reprogramming of buffalo fetal fibroblasts to induced pluripotent stem cells in feeder free culture conditions. Research in Veterinary Science, 123: 252-260. <https://doi.org/10.1016/j.rvsc.2019.01.015>
 17. Selokar NL, Sharma P, Krishna A, Kumar D, Kumar D, Saini M, Sharma A, Vijayalakshmy K, and **Yadav PS**. 2018. Establishment of a somatic cell bank for Indian buffalo breeds and assessing the suitability of the cryopreserved cells for somatic cell nuclear transfer. Cellular Reprogramming, 20 (3): 157-163. DOI: 10.1089/cell.2017.0066
 18. Vijayalakshmy K, Kumar P, Virmani M, Pawaria S, Lalaji NS, Sharma P, Rajendran R, **Yadav PS**, Kumar D. 2018. A novel combination of silane-coated silica colloid with hybrid RNA extraction protocol and RNA enrichment for downstream applications of spermatozoal RNA. Andrologia, 50(6):e13030. DOI: 10.1111/and.13030.
 19. Kumar D, Sharma P, Vijayalakshmy K, Selokar NL, Kumar P, Rajendran R and **Yadav PS**. 2018. Generation of Venus fluorochrome expressing transgenic handmade cloned buffalo embryos using Sleeping Beauty transposon. Tissue and Cell, 51: 49-55.
 20. Sharma P, Yadav AS, Selokar NL, Kumar D, Dhaka SS and **Yadav PS**. 2018. Epigenetic status of buffalo fibroblasts treated with sodium butyrate a chromatin remodeling agent. Tissue and Cell, 50: 51-58.
 21. Kumar P, Saini M, Kumar D, Bharadwaj A, **Yadav PS**. 2017. Estimation of endogenous levels of osteopontin, total antioxidant capacity and malondialdehyde in seminal plasma: Application for fertility assessment in buffalo (*Bubalus bubalis*) bulls. Reproduction in Domestic Animals; 52(2):221-226. DOI: 10.1111/rda.12882.
 22. Ghosh K, Selokar NL, Gahlawat SK, Kumar D, Kumar P, **Yadav PS**. 2016. Amnion epithelial cells of buffalo (*Bubalus Bubalis*) term placenta expressed embryonic stem cells markers and differentiated into cells of neurogenic lineage in vitro. Animal Biotechnology, 27 (1): 38-43. DOI: 10.1080/10495398.2015.1069303.
 23. Kumar D, Kumar P, Singh P, Yadav SP, **Yadav PS**. 2016. Assessment of sperm damages during different stages of cryopreservation in water buffalo by fluorescent probes. Cytotechnology, 68:451–458. DOI: 10.1007/s10616-014-9798-9.
 24. Kumar P, Saini M, Kumar D, Balhara AK, Yadav SP, Singh P, **Yadav PS** 2015. Liposome-based semen extender is suitable alternative to egg yolk-based extender for cryopreservation of buffalo (*Bubalus bubalis*) semen. Animal Reproduction Science, 159:38-45. DOI:10.1016/j.anireprosci.2015.05.010.

25. Kumar P, Kumar D, Sikka P, Singh P. 2015. Sericin supplementation improves semen freezability of buffalo bulls by minimizing oxidative stress during cryopreservation. *Animal Reproduction Science*, 152: 26–31. DOI: <http://dx.doi.org/10.1016/j.anireprosci.2014.11.015>
26. Ghosh K, Kumar R, Singh J, Gahlawat SK, Kumar D, Yadav SP, Gulati BR, **Yadav PS**. 2015. Buffalo (*Bubalus bubalis*) term amnion derived cultured cells exhibit mesenchymal stem cells characteristics in vitro. *In Vitro Cellular and Developmental Biology-Animal*, 51(9): 915-921. DOI: 10.1007/s11626-015-9920-0.
27. Mann A, Yadav RP, Singh J, Kumar D, Singh B, **Yadav PS**. 2013. Culture, characterization and differentiation of cells from buffalo (*Bubalus bubalis*) amnion. *Cytotechnology*, 65(1): 23-30. DOI 10.1007/s10616-012-9464-z
28. Singh J, Mann A, Kumar D, Duhan JS, **Yadav PS**. 2013. Cultured buffalo umbilical cord matrix cells exhibit characteristics of multipotent mesenchymal stem cells. *In Vitro Cellular and Developmental Biology-Animal*. 49 (6): 408-416. DOI 10.1007/s11626-013-9617-1.
29. **Kumar D**, Kumar P, Yadav PS. 2016. Quantitative evaluation of buffalo semen by casa during cryopreservation process. *Indian Journal of Animal Reproduction* 37 (1): 15-18.
30. Kumar D, Kumar P, Singh P, Yadav SP, Sarkar SK, Bharadwaj A, **Yadav PS**. 2014. Characteristics of frozen thawed semen in predicting the fertility of buffalo bulls. *Indian Journal of Animal Science*, 84 (4): 389–392.
31. Sadeesh EM, Balhara AK, Thirumaran SMK, Balhara S, Kumar D, Yadav S, Singh I, **Yadav PS**. 2014. Effect of hormones, sera, follicular fluid and oocyte quality on in vitro nuclear maturation and fertilization rate of Indian water buffalo (*Bubalus bubalis*) oocytes. *Journal of Cell and Tissue Research*, 14(1): 4165-4171.
32. Yadav SP, Sikka P, Kumar D, Pandey AK, Sarkar SK, **Yadav PS**, Sethi RK. 2013. Variation in milk constituents during different parity and seasons in Murrah buffaloes. *Indian Journal of Animal Science*. 83 (7): 747–751.
33. Anand T, Bera BC, Riyesh T, Vaid RK, Barua S, Virmani N, Malik P, Kumar D, **Yadav PS**, Singh RK. 2013. Cytopathogenicity of buffalopox and camelpox virus in buffalo fibroblast cells. *Indian Journal of Animal Science*, 83 (12): 1256-1260.
34. Singh P, Kumar D, Kumar P, Singh I, **Yadav PS**. 2013. Cryopreservation and quality assessment of buffalo bull semen collected from farmer's doorstep. *Agricultural Research*, 2(2):148–152.
35. **Yadav PS**, Mann A, Singh J, Kumar D, Sharma RK, Singh I. 2012. Buffalo (*Bubalus bubalis*) fetal skin derived fibroblast cells exhibit characteristics of stem cells. *Agricultural Research*, 1(2):175–182.
36. Kumar D, Mann A, Singh J, **Yadav PS**, Yadav SP, Singh P, Kumar P, Singh I. 2012. Total RNA isolation and expression of housekeeping genes in equilibrated and cryopreserved buffalo semen. *Indian Veterinary Journal*, 89 (2): 34-36.

37. Selokar NL, Sharma P, Kumar D, Kumar D, Saini M, Sharma A, Vijayalakshmy K and **Yadav PS**. 2019. Successful cloning of a superior buffalo bull Nature Scientific Reports
 38. Kumar D, Sharma P, Vijayalakshmy K, Selokar NL, Kumar P, Rajendran R, **Yadav PS**. 2018. Generation of Venus fluorochrome Expressing Transgenic Handmade Cloned Buffalo Embryos Using Sleeping Beauty Transposon. *Tissue Cell*. 51:49-55.
 39. Sadeesh EM NL Selokar , AK Balhara and Yadav PS. 2016 Differences in developmental competence and gene expression profiles between buffalo preimplantation mebyros cultured in three different media Cytotechnology Cytotechnology 68(5):1973-86. DOI 10.1007/s10616-016-0010-2
 40. Baldev R Gulati, Rajesh Kumar, Niharika Mohanty, Sandeep Gera, Pawan Kumar and Prem S Yadav 2016 Comparative evaluation of equine mesenchymal stem cells derived from amniotic fluid and umbilical cord blood Indian Journal of Animal Sciences 86 (5): 539–544.
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- International Conference on Biotechnology, Department of Biotechnology, CDLU Sirsa. Sept.18-20, 2012 pp195 (Best poster award)
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 42. **Yadav, P.S.**, S. Khanna, O.K. Hooda and S.K. Jindal (1999). Role of heterologous oviductal cell co-culture in cleavage and development of Buffalo IVF embryos. 15th Annual Conference of ISSAR, PAU, Ludhiana, Feb. 9-11.
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50. D.V Arneja, **P.S. Yadav,** G.C. Georgie and M.L. Kaker (1989). Thyroid weight, plasma triiodothyronine and thyroxine concentration in Japanese quails as affected by age, energy and vitamin concentration of ration. 5th Annual Conference of SAPI, Kerala Agriculture University, Dec. 27-30.
51. **Yadav, P.S.,** D.V. Arneja, G.C. Georgie and I.S. Lohan (1989). Testicular size and plasma testosterone concentration in Japanese Quails as affected by age, energy and vitamin concentration of ration. 5th Annual Conference of SAPI, Kerala Agriculture University, Dec. 27-30.

Conferences Attended

International

1. International Conference on Biotechnology September, Department of Biotechnology CDLU Sirsa. 18-20, 2012
2. International Buffalo Conference at NASC Complex Delhi Feb 1-4 2010
3. International Conference "Germ Cell Development Function and 19th Annual Meet of ISSRF January 22-24.2009 at IISc Bangalore.
4. Ist International Stem Cell Submit 2008, IIT, Chennai, Nov 14-16 2008.
5. International Symposium on Stem Cells Premises & promises for Research & Therapeutics, Mumbai 18-21 Sept 2005
6. International meeting Stem Cell network North Rhine Westphalia Bonn Germany. April 1-2 2004
7. Asian Buffalo Congress Surya Hotel Delhi February 25-28 2003
8. International Conference Biotechnology in Animal Reproduction at Chennai December 2-4, 2002.
9. Asian Australasian Society of Animal Production September Ashoka Hotel Delhi 23-37, 2002.

10. International Symposium on Sustainable Animal Production , Health and Environment and future challenges for milk meat and Draft At CCS HAU 24-27 Nov.1999

National

1. Society of Animal Physiologists of India, Mathura November 19-21 2013
2. Society of Animal Physiologists of India, November 6-8 Navsari 2012
3. National Conference of Medial Biotechnology, April 16-18, MDU Rohtak
4. Annual Conference of SOCDAB 2010 Feb 10-11 Anand, Gujrat
5. XVth Annual Convention of Indian Society of Veterinary Immunology and Biotechnology Feb 26-28, 2009, CCSHAU Hisar
6. IX th Annual Conference of Indian Society of Animal Genetics and breeding NASC Complex Delhi July 3-4 2008.
7. Participated in 7th scientist meet of network project on buffalo improvement April 4-5, 2007 at CIRB Hisar
8. Federation of Indian Physiological Societies, Delhi Jan 11-13, 2007
9. Indian Society for Study of Reproduction and Fertility NDRI Karnal. Feb 23-25 2006.
10. Indian Society for Study of Animal Reproduction at SKUAST R S Pura Campus Jammu Nov. 23-25, 2005.
11. Annual conference of SOCDABNBAGR Karnal, Feb 10-11 2005.
12. Annual conference of Society of Animal Physiologists of India at Durg, Chhattisgarh, Jan. 28-29, 2005.
13. Indian Society for Study of Animal Reproduction at IVRI Izatnagar Nov. 14-16 2002.
14. Society of Animal Physiologists of India, Palampur, Oct. 18-19, 2001.
15. Indian Society for Study of Animal Reproduction Jodhpur, Oct. 6-8, 2001.
16. Federation of Indian Physiological Societies Calcutta Nov. 24-27. 2000.
17. National Seminar for Buffalo Development CCSHAU Hisar 14-16 Oct 1999.
18. Indian Society for Study of Animal Reproduction at PAU Ludhiana 9-11 Feb, 1999.
19. Indian Society for Study of Animal Reproduction at GBPUA&T Pantnagar 9-11 Feb , 1996.
20. Annual conference of Society of Animal Physiologists of India 1995 at CARI, Izatnagar
21. Annual Conference of Society of Animal Physiologists of India 1992 at CCSHAU Hisar
22. Annual Conference of Society of Animal Physiologists of India 1989 at Trichur, Kerla